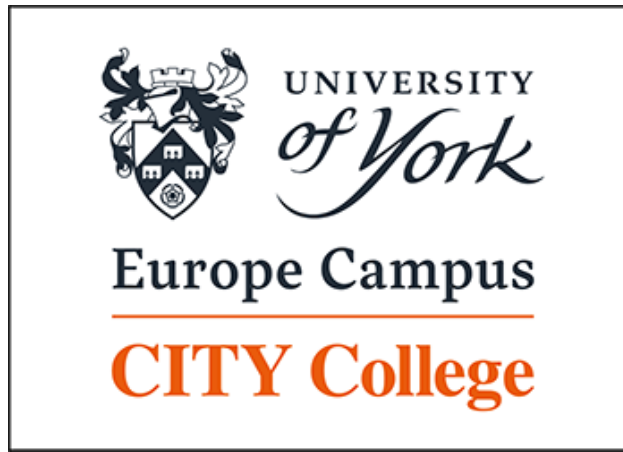




Computer Science Department

Undergraduate Individual Project Handbook

**Guidelines & Regulations
2025-2026**

BSc in COMPUTER SCIENCE

BSc in COMPUTER SCIENCE (Web Technologies)

BSc in COMPUTER SCIENCE (Business Informatics)

**BSc in COMPUTER SCIENCE (Artificial Intelligence and
Data Science)**

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1. Introduction

The purpose of this document is to present guidelines and regulations for the **Undergraduate Capstone Project Module (CPM)** and preparing its written Dissertation for the following programmes:

- BSc in COMPUTER SCIENCE
- BSc in COMPUTER SCIENCE (Web Technologies)
- BSc in COMPUTER SCIENCE (Business Informatics)
- BSc in COMPUTER SCIENCE (Artificial Intelligence and Data Science)

Read the Handbook carefully and keep it throughout your final year for future reference.

1.1 Description

The undergraduate individual project is an extended piece of individual work, occupying your time throughout Stage 3 of your undergraduate studies. You will work on a topic that interests you and have regular meetings with your supervisor to discuss your work.

It involves two aspects:

- The work you do and the processes you follow towards solving a problem, leading to the development of a software product.
- The written report, formally known as dissertation, you submit at the end, reporting all your work, findings, and results.

1.2 Aim and Objectives

Writing a dissertation or undertaking a project provides students (including undergraduates and taught postgraduates) with the opportunity to undertake a piece of individual research/investigation and examine an aspect of the subject they have been studying in more depth. Such tasks can therefore assess students' ability to:

- work independently;
- narrow / define / focus a research area of their choice;
- read widely and critically reflect on written research in an appropriate and thorough manner;
- think through varying methodological approaches and adopt the necessary approaches suitable to the topic being researched;
- conduct research;
- manage a challenging, extended piece of work.

The project will develop your ability to:

- construct a project from initial unstructured ideas, through a thorough analysis of the problem area;
- use libraries and other information sources;
- apply tools and techniques from taught courses;
- demonstrate practical skills and understanding of theory;

- apply theoretical concepts to practical problems;
- plan, schedule, monitor and control own work over a long period of time;
- communicate findings in a variety of ways e.g. written report and oral examination; and
- defend your ideas in discussions and presentations.

1.3 The CPM in relation to your Degree

The project will appear on your transcript as module **CCS3900 – Individual Project** and is equivalent to **40 credits**.

1.4 The CPM in relation to your Track

It is a formal requirement that the project reflects the skills and knowledge you acquire depending on the track you have selected when you progressed from Stage 2 to Stage 3. Failure in this may result in a significant penalty on your dissertation mark.

More specifically, students of all tracks should clearly demonstrate that they have developed a thorough comprehension of computer systems as a whole, with a focus on the issues, problems, and processes associated with the analysis, design, implementation, and testing of systems.

In addition to the above, a student in the **Web Technologies** track should concentrate on a significant software component related to the Internet and should clearly demonstrate that they have developed a thorough understanding of the main principles and have acquired expertise on technologies relating to internet computing, (including security, distributed systems, networking), together with the appropriate skills for the development of such software systems. In a Web Technologies project, the emphasis is on the use of the best available technologies that best suit the problem at hand.

Likewise, a student in the **Business Informatics** track should clearly demonstrate that they have acquired a mixture of technical and contextual knowledge and skills, appropriate both for the development of software systems and for the strategic deployment of current and emerging technologies, for promoting innovation and entrepreneurial activities within a modern organisation. More particularly, a Business Informatics project, apart from software development, should focus on the business-added value and aspects of the project. Because the emphasis is on creating and delivering a solution rather than exploring cutting-edge technologies, such projects may imply the use of existing and well-versed technologies.

Finally, a student in the **Artificial Intelligence and Data Science** track should clearly demonstrate that they have acquired a mixture of technical and contextual knowledge and skills, appropriate for the development of intelligent systems, typically capable of learning from data. Emphasis is again placed on the process of developing such systems since the steps involved and their lifecycle is somewhat different.

Having regular meetings with your supervisor will help you guide your dissertation to reflect your track.

1.5 Workload

Given that (i) 1 credit is equivalent to 10 hours of work, (ii) the project is worth 40 credits, and (iii) the project lasts all year, you are expected to work on your project at least 15 hours per week during the Fall and Spring semesters of Stage 3, which amounts to **400 working hours**.

1.6 Supervision and Feedback

Students are entitled to reasonable access to their supervisors. The total number of meetings, their frequency, and their duration, may vary depending on the student's progress, their individual needs as well as the stage their dissertation project has reached.

UG students of the CSD are typically **expected to meet their supervisor bi-weekly** (during the teaching weeks of the academic year).

- Responsibility for arranging supervision meetings lies with the student. To arrange them, students are advised to agree on and pre-schedule regular meetings as recurring events (rather than ad-hoc).
- You are also strongly encouraged to keep a *logbook* to record what was delivered, discussed, and decided in each of these additional meetings. The logbook may vary in format but a Google Form has been created specifically for this purpose (link available on the VLE). Students are expected to complete this form after each weekly meeting with your supervisor. Every log entry is sent by email to both you and your supervisor.

UG students of the CSD are also **formally required to attend three supervision meetings** at different stages of the dissertation process.

- For UG Stage 3 students of the CSD, these will typically (but not necessarily) take place:
 - After the Portfolio Submission
 - Before the Software Demonstration
 - Before the Dissertation Submission
- Responsibility for arranging these three supervision meetings lies with the supervisor.
- Responsibility for keeping logs for these three meetings lies with the supervisor. The supervisor should share the meeting log with the student who is required to respond only in case they disagree with the content.

With regard to **feedback**, supervisors are expected to:

- Provide oral feedback on any work produced by the student and presented during a supervision meeting. Supervisors may also ask you to submit short reports on your work, recording the main milestones of the project, and discussing both progress as well as difficulties.
- Review a draft of the entire dissertation, provided that students have met the agreed deadlines and submitted their work on time.

Feedback on the entire dissertation should normally be provided within 10-15 working days. However, feedback times may differ depending on the student's progress, adherence to deadlines, and consultation with the supervisor. Feedback on the entire dissertation can be in written or oral form, but must be recorded when given verbally (e.g. if the meeting is online, the lecturer may record it).

Students are not allowed to submit the same piece of work for feedback to their supervisor more than once.

1.7 Backing up your work

Most supervisors will request that you allow them access to your work, throughout the duration of the CPM, for easy access and feedback purposes. This also serves as an effective **backup** strategy. As such, you are strongly encouraged to:

- store your work (dissertation draft, resources, and other documents) in a shared (with your supervisor) folder on a cloud platform, such as Google Drive; and
- use a **Git service** (e.g. GitHub) for your software's source code.

It is vital that you understand that **you are solely responsible for not losing your work**.

1.8 Ethics and code of conduct

The paramount principle governing all research at the University of York involving human participants and personal data is respect for the participants' welfare, dignity, and rights. To this end, the University of York has established an Ethics approval process that you, as the principal investigator of your research, have to follow. This process involves the application for approval of research, whenever third parties are directly involved in your work/research. For more details on the ethics approval process, you should talk to your supervisor and attend the ethics training session that will take place at the beginning of your project.

As a future computer scientist and graduate of a BSc programme that is accredited by the British Computer Science (BCS), you are expected to abide by the rules that apply to any IT professional, as these are described in the BCS Code of Conduct, which is available online at: <https://www.bcs.org/category/6030>.

1.9 Use of Unfair Means

You are required to submit your own original work. Where other material is used, you must state the sources from which the information is derived and the extent to which you have availed yourself of the work of others. Failure to acknowledge the sources used may be interpreted as plagiarism.

In addition, any use of Generative AI for any part of the dissertation project (either in writing the dissertation or code) is strictly prohibited. Any piece of work in which plagiarism is clearly identifiable will normally be given a fail grade. Any incident of use of unfair means is formally recorded and may result in disciplinary action.

Taking into consideration all the above, when submitting the dissertation you will be asked to sign the following **declaration**:

- All sentences or passages quoted in this dissertation from other people's work have been specifically acknowledged by clear cross-referencing to author, work and page(s). I understand that failure to do this amounts to plagiarism and will be considered grounds for failure in this dissertation and the degree examination as a whole.
- I have completed and submitted this work by myself without assistance from or communication with another person either external or fellow student or any AI type of content generator. I understand that not working on my own will be considered grounds for unfair means and will result in a fail mark for this work and might invoke disciplinary actions.

1.10 Troubleshooting

It is imperative that if, at any point throughout this period, you are facing problems that are affecting your work on the project, you contact one of the following as soon as possible (a) your project supervisor; (b) the Academic Director of Undergraduate Studies; or (c) the Head of the Department so that they provide you with the necessary help/support.

2. Process Outline and Timetable

2.1 Stage 2 Process Outline and Timetable

In the previous stage of your studies you were given a presentation and were invited to “Tracks and Individual Project Preparation” (on your VLE) so as to have all necessary information about the process of selecting a Project for Stage 3 of your studies.

2.1.1 Project Proposals, mid Stage 2 Fall Semester

You have received a list of proposed projects and their description. You were given some time to meet the faculty members who have submitted proposals in order to express your interest as well as to discuss the proposed topics that interest you.

You were also encouraged to propose your own project proposal. If you have an idea you could discuss it with the most relevant member of the staff so that they help you to formulate your idea in a proposal.

2.1.2 Project Topic Choices Submission, end of Stage 2 Spring Semester

The dissertation choice form, which was available on “Tracks and Individual Project Preparation” on your VLE, was completed to record your preferences.

2.1.3 Project Allocation, end of Stage 2 Academic Year

You have been allocated a project and a supervisor. The list of project allocations was announced on “Tracks and Individual Project Preparation” on your VLE and you were advised to contact your supervisor.

2.2 Stage 3 Process Outline and Timetable

All the following dates are **provisional**. Should there be any changes, you will be notified in due time through “CCS3900 Individual Project” on your VLE.

2.2.1 Project Portfolio, Stage 3 Fall Semester, Week 4

You are required to submit a portfolio of initial project work to your supervisor by the end of week 4. This portfolio sets the scene for your project. The contents of the portfolio will be determined by your supervisor, but at minimum, they should contain the following:

- A short description of the project (~200-300 words).
- The aim of the project, briefly but clearly stated (~50 words).
- A short rationale/motivation of the project (why it is interesting, who will benefit from it, ~200-300 words).
- A short discussion of any background issues involved (~150 words)
- A short argumentation about how the project fits with the track you selected (~150 words).
- A short plan of action for the rest of the semester, clearly identifying tasks and their completion time (GAANT chart).
- A past dissertation assessment.

After the portfolio submission, in a discussion with your supervisor, the viability of the project will be considered.

2.2.2 Progress Review Panel, Stage 3 Fall Semester, Week 15

You will appear in front of a panel of 2-3 academic staff members to present your project, your overall progress in relation to the plan you submitted with the portfolio, and a self-evaluation of this progress. It is expected that the project will be well under way at this stage and a detailed plan up until the dissertation submission date should be presented. Students in the Web Technologies, Business Informatics, and Artificial Intelligence and Data Science tracks, you will also need to discuss with the panel why and how you believe this work is relevant to the track you selected.

The interview will last approximately 15 minutes.

2.2.3 Software Demonstration, Stage 3 Spring Semester, Week 8

This is a critical milestone in the development of your project as at this stage you are expected to demonstrate to the academic staff the main software product. Although the software product is not expected to be fully complete at this point, it should be nearing completion and you should be able to demonstrate its main functionalities. Academic staff will give you feedback on your progress, your oral communication skills, and in relation to the aim and objectives that were originally set in the portfolio.

2.2.4 Dissertation Submission, Stage 3 Spring Semester, Week 12

When you have written the dissertation and have the approval of your supervisor to submit you will need to submit the following:

- two spiral-bound double-sided printed copies of your final dissertation to the Course Administrator.
- electronically using a Google form (available on the Google classroom):
 - Your dissertation in .pdf format.
 - Your dissertation source files (.doc, .docx, .tex, etc.)
 - A video presentation of your project (.avi or .mp4), lasting no more than 3 minutes, demonstrating the developed application. You may use one of the many screen-capturing software applications (desktop or online) that are freely available.
- The source code of your software product on GitHub.
- An electronic version of your dissertation to the TurnItIn plagiarism detection software.

2.2.5 Viva (Oral examination), Stage 3 Spring Semester, Week 15

As a result of your submission, you are required to attend a viva, which typically consists of you, your supervisor, and your 2nd Examiner; in some cases, an additional academic staff member may also be present as Chair. The viva typically takes place within two weeks after your submission.

During the viva, which lasts ~30 minutes, you will be expected to demonstrate your final software product, defend your work, state main achievements, discuss shortcomings, and respond to questions set forth by the 2nd Examiner.

The oral examination is an opportunity for you to demonstrate that:

- the work is your own,
- you understand what you have done,
- you can explain how the project was developed,
- you can clearly explain why particular tools were used amongst those available,
- you have read widely,
- you can accept the project shortcomings and discuss its achievements, and
- you can show an understanding of how it could be improved in the future.

After the viva, the supervisor and the 2nd Examiner will decide whether the student has successfully completed the undertaken project or failed to reach the minimum standards required. It is also possible that they ask you to make minor or major amendments to your dissertation before submitting the final copy.

3. Assessment

Assessment of your CPM CCS3900 - Individual Project module is *exclusively based on the dissertation* you submit at the end of the project. However, should you not submit/attend any of the previous non-assessed deliverables (portfolio, progress panel, and demonstration), the examination board may take this into consideration.

3.1 Dissertation Assessment Criteria

The dissertation is assessed *independently* by your supervisor and your 2nd Examiner based on the following assessment criteria:

Problem description – Aim – Objectives (15%)

- Motivation is clearly presented in the introduction.
- The aim of the project is clearly stated in the introduction.
- The objectives of the specific project (and not of the student, or any project) are clearly presented in the introduction and are relevant to the aim.
- The problem is described in different levels of abstraction in various sections inside the report.

Literature review and Use of Resources (20%)

- Appropriate related work (references) is identified and presented.
- Appropriate background theory related to the project is presented and analysed in appropriate depth so that the reader is able to read through and understand the remaining report.
- Literature review has a structure that demonstrates synthesis skills and is not merely a listing of other people's work. The student summarises and concludes with points relevant to their work. All topics directly related to the project are presented in the right context.
- The literature review is comprehensive and up-to-date.
- All sources are correctly cited within the text and correctly listed in the References Section.
- Appropriate tools have been identified.

Problem-solving process (25%)

- The student has identified the tasks necessary for fulfilling the aim of the project.
- A problem-solving and/or software engineering process is proposed.
- The process proposed is appropriate and justified.
- There is evidence that the student followed the proposed process.
- Each stage of the process is reported in appropriate length.
- The methods employed for the analysis, design, implementation, testing and evaluation are appropriate for the project.

Products (25%)

- Every product that is expected according to the process followed is clearly presented in the report.
- The methods employed for the analysis, design, implementation, testing and evaluation have been implemented correctly.
- The developed products are of high quality.

Evaluation (5%)

- The objectives of the project have been met.
- The student identifies the strong and weak aspects of their work.
- The student evaluates the process that has been followed.
- The results of the project have been related to other peoples' results.
- The student suggests directions for further improvement.

Clarity of Communication (10%)

- General Layout
- Dissertation write-up guidelines have been followed.
- General Report Structure and Flow
- The report is sensibly structured into chapters, sections, paragraphs, etc.
- All chapters/sections have appropriate titles.
- There are appropriate tables, figures, code snippets, etc, supporting the text.
- Use of language
- There are no spelling, grammar, or syntax mistakes.
- Use of formal language, appropriate for academic purposes.

3.2 Compensation and Reassessment

As a module, the CPM is:

- **NOT compensatable**
- **re-assessable**, in the case of a **marginal fail**, i.e. a mark in the range 30-39. The student should be able to undertake the work required to bring this up to pass level:
 - without access to the University's physical facilities;
 - without further supervision;
 - within two weeks full-time equivalent effort.

When re-submitting their dissertation students will be required to include a cover sheet detailing the changes they have made.

The final dissertation mark after resubmission in this case will be capped at 40.

In the case of an **outright fail**, i.e. a mark in the range 1-29, there is no opportunity for reassessment and the student will not be eligible for a Bachelor's Degree.

3.3 Extenuating Circumstances

3.3.1 Extensions to the Deadline

Extensions for the dissertation can be granted for a **maximum of 15 days** from the date of submission and only **due to exceptional circumstances**, i.e. medical reasons (medical certificate must be provided) or serious personal issues.

Note that it is not within the power of the supervisor, the Academic Director, or the Dissertation Coordinator to grant extensions; extension requests are reviewed by the Department. If you are experiencing problems with your dissertation you must inform your supervisor immediately for advice.

Students must apply for an extension due to extenuating circumstances **at least three weeks before** the submission date by filling in the “Extenuating Circumstances Form” available on the VLE. If you apply for an extension, you are most strongly advised to continue working on your dissertation while your request is being considered. Note that **extensions cannot be applied for and granted retrospectively**.

For further information about the procedure of applying for an extension and the supporting documentation, please consult the “Extenuating Circumstances” section of the Student Handbook and your Student Corner.

3.3.2 Leave of Absence (Inactivity)

In cases that the effort of the student is severely compromised due to lengthy extenuating circumstances, the student is strongly advised to request a leave of absence (inactivity) for a maximum of one year. If circumstances allow the student to become active again or the inactivity period is exhausted, the student must then submit on the next appropriate submission date.

Dissertation submission dates throughout the year are:

- August 31st
- January 31st
- May 31st

3.4 Failure to submit the dissertation

When a student fails to submit the dissertation by the deadline, they automatically fail the dissertation. Since there is no re-assessment option, the student is not in a position to graduate. They might be, under particular circumstances, eligible to be awarded an ordinary (non-Honours) degree.

4. Dissertation Write-Up Guidelines

The dissertation must be in English and typewritten using a word processor or LaTeX. All printed submissions must be of good quality, making sure that all text and figures are readable.

4.1 Dissertation Length

An undergraduate dissertation (main body, i.e. excluding preamble, references, and appendices) is expected to not exceed 10,000 words or 50 pages in length.

4.2 Dissertation Template

A template will be provided to you on the Google Classroom of the Individual Project in two different formats:

- Word document (.docx)
- LaTeX

You are **required** to use one of these two. Both of them adhere to the following guidelines.

4.2.1 Dissertation Structure

Typically a dissertation includes the following:

The preamble

Title Page: Each dissertation must start with a title page, laid out in a similar way to that provided in the template (see Section [4.2 Dissertation Template](#)).

Abstract: An abstract of the dissertation should be provided with no less than 150 and not exceeding 250 words. The Abstract page should appear right after the title page.

Declaration: On the following page, the dissertation must include a declaration (see Section [1.9 Use of Unfair Means](#)).

Acknowledgements: This is an optional page where you may express your appreciation and gratitude to people who have assisted you in preparing the dissertation.

Table of Contents: Include page numbers of all sections.

Tables of Figures

Table of Tables

Use roman numerals (i, ii, iii, iv, etc.) for all pages of the preamble.

The main body of the report

- Chapter 1: Introduction (begins on page 1).
- Chapter 2: Background Work / Literature Review.
- Chapter 3...n: The main achievements of the project (see below).
- Chapter n+1: Results, Critical discussion and Evaluation of the results of the project.
- Chapter n+2: Conclusions and Future Work.

- References: The References section must be placed after the last chapter of the report and before the appendices. The proposed citation style is that of IEEE (see Appendix A) but you may use another as long as you are consistent.

The Appendices

Possible appendices may include:

- Diary of when major milestones were achieved
- Additional figures/diagrams.
- Tables of data (if appropriate).
- Illustrative parts of the software code.
- A user manual or system documentation (if appropriate).

4.2.2 General Formatting

The main body of the dissertation should be single-spaced and set to a 12-point Times New Roman typeface font. Headers and footers should be set to a 10-point typeface.

Chapters start on a new page and their titles should be set in a larger font than the text. They must be divided into numbered sections and then into paragraphs. For paragraph formatting, make sure there is a 6pt spacing after.

Equations, theorems, tables, and figures must be numbered appropriately, i.e. figures in chapter 1 should be numbered Fig. 1.1, Fig. 1.2, Fig. 1.3, ..., figures in chapter 2 would be numbered Fig. 2.1, Fig. 2.2, Fig. 2.3, ..., equations in chapter 3 would be numbered Eq. 3.1, Eq. 3.2, Eq. 3.3, ..., and so forth.

Appendices should be named by single letters (A, B, C, etc.).

4.2.3 Margins and Paging

The whole document (body, title page, tables of contents, etc.) should have 30mm margins on the top/bottom/left/right sides of the page.

A header on each page should state the chapter number and title and should be right-aligned. A footer on each page should indicate the page number, right-aligned. Headers and footers should be placed at least 2cm from the top/bottom of the page respectively.

The Table of Contents, Table of Figures, and Table of Tables should be placed in the preamble of the document and their pages should be numbered with roman numerals (i, ii, iii, etc.). The numbering of the pages of the main body should be in Arabic numerals with numbering starting on the first page of the first chapter.

Pages in the appendices should be numbered with the appendix letter and an Arabic numeral (A-1, A-2, etc.). Numbering should restart on the first page of each Appendix.

4.3 Writing the dissertation

It is suggested that you adopt the following strategy:

- Produce a plan of the dissertation, showing the division into chapters and sections, the headings for each chapter and section, and possibly a sentence or two indicating how you intend to cover the material in each chapter or section. You should then discuss this plan with your supervisor. Do this as early as possible, ideally at the start of the Spring semester.

- For each chapter or section, you should produce a first draft, and discuss this with your supervisor. You will probably find it convenient to do this chapter by chapter.
- In light of your supervisor's comments on the first draft, you should produce the final version.

The introduction chapter serves to introduce the reader to the wider topic of the project, clearly outline the motivation, and state the aim and the objectives of the project. It is usual to assume that the reader is familiar with the fundamentals of Computer Science. Any further material should be appropriately presented to the reader and supported by suitable references. The introduction sets the scene for the project. Grab the reader's interest early.

A review of relevant work by other authors and the relationship between this and your own work is typically included in Chapter 2. If several other people have done closely-related work in a different way, then the reasons for your approach should be summarised here.

The structure of the main chapters will depend on the nature of the project and the selected problem-solving/software development process. You should include chapters on the requirements analysis, specification, design, implementation and testing phases of the development. It is suggested that the method for verification/testing of the software is considered both at the specification as well as the design stages so that program design and possible testing strategies are developed together. In the design chapter, it is often a good idea to start by discussing why you chose the particular design technique(s) from those available, before giving an overview of your design (typically a diagram) and then proceeding to details.

In any case, it is important that you acknowledge the work of others that you use in your own work. The use of references is the standard way to do this. You should also attempt to use material and techniques from courses taught throughout the duration of your studies, e.g. design techniques, project management tools, etc.

The Evaluation chapter of the dissertation should first present the main results of your work and then critically discuss these. Discuss whether the objectives have been met and justify your opinion. It is important that any failure to achieve particular objectives should be discussed. This does not mean that you will be penalised. One fundamental aspect of the assessment procedure is that the processes (how you went about your project) are as important as the products of the project. You must be able to critically evaluate your own work.

The Conclusions chapter can be a fairly short chapter (2 or 3 pages) that summarises your work and brings together the points that you have made in previous chapters. Do not be afraid of repeating some of your earlier statements here. This chapter should also present future work that you feel would be of interest or should be carried out on the work that you have done.

You are advised that writing a dissertation is a fairly time-consuming process and it is in your interests to make an early start on it. Your supervisor will probably discuss a timetable with you for each stage of the work and you are strongly recommended to try to stick to it. More specifically, you are advised to produce the first drafts for some chapters before all work has been completed and not wait until you have done all the practical work before starting to write the dissertation. For example, Chapter 2 can usually be written first. Maintain your list of references as you work rather than try and assemble it at the end.

Finally, always make sure you **spell-check and proofread** your work. In relation to structure and formatting, make sure you consult the **Report Formatting Checklist** that has been provided to you since Stage 1 on your Student Corner. Your supervisor is not a copy editor; they will

comment on structure and content but will not correct a multitude of grammar and spelling mistakes.

A few final hints:

- Do not write in the first person singular. Passive voice is more formal so “A program has been developed...” is preferable to “I wrote a program...”.
- Avoid contractions. (“Do not” instead of “don’t”).
- A reader who will not read the entire dissertation (e.g. the Chair of the oral examination, if one is appointed) should be in a position to grasp a fair idea of the work you have done simply by reading the abstract, introduction and conclusions so make sure that they are informative and to the point.
- Do not include a printout of all the source code! Some snippets of code may be included in either the main body of the report or in an Appendix, only if they are relevant to the discussion.

Revised: 2025 October

Dr Ioanna Stamatopoulou, Head of Computer Science Department

Appendix A: IEEE Citation Style

References in the text must match the reference list both in number and style. All sources must be mentioned in the text. For more information, check the [IEEE Citation Guidelines](#) or visit this [practical guide](#) on the University of York website.

Note that referencing systems other than IEEE (e.g. Harvard - [practical guide](#) on the University of York website) are also acceptable as long as they are used correctly and consistently.

Citations within the Text

- References must be numbered in the order in which they appear in the text.
- Once you label the source, use the same number in all subsequent references.
- Each reference number should be enclosed by square brackets on the text line, with a space before the bracket, and before the punctuation: "... end of the line for my research [12]."
- It is not necessary to mention the author(s) of the reference unless it is relevant to your text. Do not mention the date of the reference in the text.
- It is not necessary to say "in reference [27]. . . ." "In [27] . . ." is sufficient.
- To cite more than one source at a time:

Vastly preferred

[1], [3], [5]

[1] - [5]

Acceptable

[1, 5, 7]

[1-5]

- The names of all authors should be given in the references unless the number of authors is greater than six. If there are more than six authors, you may use *et al.* (no comma before *et*) after the name of the first author.

The Reference List

- References must be listed in the order they were cited (numerical order), not in alphabetical order.
- The bracketed number should be on the first line and all following lines of each entry indented.
- List only one reference per bracketed number.
- Footnotes should not be included in the reference list.

Capitalisation

- Every (important) word in the title of a *book* must be capitalized.
- Every (important) word in the title of a *journal* or *conference* must be capitalized.
- Capitalize only the first word of an *article* title (except for proper nouns, acronyms, etc.)
- Capitalize only the first word of a *paper*, *thesis*, or *book chapter*.
- Capitalize the "v" in *volume* for a book title, but not for a periodical.

Abbreviations

- You must *either* spell out the entire name of each periodical you reference or use accepted abbreviations. You must consistently do one or the other.
- You may spell words such as *volume*, *December*, etc., but you must either spell out all such occurrences or abbreviate all.
- You do not need to abbreviate March, May, June, or July.
- To indicate a *page range*: pp. 111-222, but to reference *one page*, use only one *p*: p. 111.

Spacing

- Please note the correct spacing and punctuation for author names:

D. L. Tao, C. Siva Ram Murthy, and S. Al Kuran
but

T.-C. Hsu and L. A. Stein-Rosenberg

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- [1] S. M. Hemmingsen, *Soft Science*. Saskatoon: University of Saskatchewan Press, 1997.
- [2] A. Rezi and M. Allam, "Techniques in array processing by means of transformations," in *Control and Dynamic Systems*, Vol. 69, *Multidimensional Systems*, C. T. Leondes, Ed. San Diego: Academic Press, 1995, pp. 133-180.
- [3] D. Sarunyagate, Ed., *Lasers*. New York: McGraw-Hill, 1996.

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- [3] G. Liu, K. Y. Lee, and H. F. Jordan, "TDM and TWDM de Bruijn networks and shufflenets for optical communications," *IEEE Transactions on Computers*, vol. 46, pp. 695-701, June 1997.
- [4] J. R. Beveridge and E. M. Riseman, "How easy is matching 2D line models using local search?" *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 19, pp. 564-579, June 1997.

Articles from Published Conference Proceedings

The word *in* before the conference title is not italicized.

- [5] N. Osifchin and G. Vau, "Power considerations for the modernization of telecommunications in Central and Eastern European and former Soviet Union (CEE/FSU) countries," in *Second International Telecommunications Energy Special Conference*, 1997, pp. 9-16.
- [6] S. Al Kuran, "The prospects for GaAs MESFET technology in dc-ac voltage conversion," in *Proceedings of the Fourth Annual Portable Design Conference*, 1997, pp. 137-142.

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- [7] H. A. Nimr, "Defuzzification of the outputs of fuzzy controllers," presented at the 5th International Conference on Fuzzy Systems, Cairo, Egypt, 1996.

Reports (technical reports, internal reports, memoranda)

Provide number and month if available.

- [8] K. E. Elliott and C. M. Greene, "A local adaptive protocol," Argonne National Laboratory, Argonne, France, Tech. Rep. 916-1010-BB, 1997.

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"PhD dissertation", but "MSc thesis".

- [9] H. Zhang, "Delay-insensitive networks," M.S. thesis, University of Waterloo, Waterloo, ON, Canada, 1997.

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- [11] T. I. Wein (private communication), 1997.

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- [12] Computational, Optical, and Discharge Physics Group, University of Illinois at Urbana-Champaign, "Hybrid plasma equipment model: Inductively coupled plasma reactive ion etching reactors," December 1995, <http://uigelz.ece.uiuc.edu/Projects/HPEM-ICP/index.html>.
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